

OM RAHEJA

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Education

University of Illinois Urbana-Champaign (UIUC)

Expected May 2029

Bachelor of Science in Computer Engineering

4.0/4.0 GPA

Technical Skills

Languages: C, C++, x86 Assembly, Python, Go, SQL, JavaScript, HTML/CSS

Hardware & Systems: QEMU, Linux Internals, OpenSCAD, OnShape, KiCad, SKiDL, I2C, Microcontrollers (ATtiny)

Frameworks & AI: React, Next.js, Node.js, Express, TensorFlow.js, PostgreSQL, Agentic AI, VLMs

Developer Tools: Git, Google Cloud Platform, Linux CLI

Work Experience

codeabode | *Python, Java, 3JS, Cloudflare Workers, Next.js*

April 2024 – Present

Chief Executive Officer & Lead Developer

Plainsboro, NJ

- Coordinated a 7-person team to deliver 70+ hours of instruction to 100+ students, generating \$2,000 in revenue
- Architected a Python AI pipeline to dynamically generate personalized programming curricula based on real-time student performance metrics
- Designed an interactive 3D game with three.js to visually simulate core programming paradigms for students

AIOS – Rutgers University | *Python, LiteLLM, AgentBench, LaTeX*

May 2024 – Dec 2025

Research Assistant

New Brunswick, NJ

- Co-authored AutoFlow, an automatic agentic workflow generator optimizing LLM task execution and coordination
- Detailed open-source documentation covering framework installation, system usage, and contributor guidelines
- Lead speaker presenting to 50+ students & PhDs on agentic infrastructure architectures at the AIOS Hackathon

aquarc | *Selenium, Go, React, PostgreSQL, VectorDB, Google Analytics*

June 2024 – July 2025

Chief Executive Officer & Lead Architect

Remote

- Innovated an enterprise SAT prep platform scaling to 8,000+ questions using Go, React, and PostgreSQL
- Delegated tasks within a remote team of 10+ developers from product conception through agile deployment cycles
- Drove growth to 1,000+ active users by leveraging Google Analytics and launching a targeted beehiiv newsletter

Project Experience

MilkBoiler | *ATtiny1614, I2C, MOSFET, UPDI, Python, SKiDL, KiCad 10, OpenSCAD*

Sep 2026

- Engineered an automated stove shutoff system that detects milk boil-over to prevent spills and fire hazards
- Architected an ATtiny1614-based embedded system incorporating I2C sensor integration, MOSFET-driven servo actuators, and UPDI programming
- Automated PCB netlist generation using the Python SKiDL framework for KiCad 10, and modeled a parametric OpenSCAD 3D enclosure with thermal ventilation

SwimIQ (swim-iq.vercel.app) | *React, TensorFlow.js, Express, Groq Llama 4 Scout*

July 2026

- Engineered a premier swim-coaching app to help coaches and athletes analyze and improve technique
- Built an in-browser computer vision pipeline using TensorFlow.js and MoveNet Thunder to track 17 keypoints at 30 FPS, applying custom kinematics to filter out water glare
- Designed a hybrid edge/cloud architecture utilizing Groq Llama 4 Scout vision model to generate real-time AI feedback

MeetTimeTracker | *Python, Gemini API, OCR, JSON*

Jan 2026

- Accelerated swim coach data entry by 20x, streamlining paper-to-digital result tracking for online platforms
- Engineered an AI agent leveraging Gemini OCR to parse physical swim meet papers into a centralized database
- Formulated custom prompt algorithms to resolve text overlap and row alignment errors, automating data conversion into structured JSON payloads

omOS | *C, x86 Assembly, QEMU*

Feb 2024

- Wrote a standalone C standard library and a VGA graphics driver integrated with a hardware interrupt keyboard handler
- Utilized QEMU for system emulation, kernel debugging, and low-level memory mapping

Leadership & Advocacy

Legislative Advocacy & AIxEducation | *National Finalist & Co-Author*

Sep 2024 – Nov 2024

- Won 2nd Place Nationally for AI research simulating collaborative Montessori schools, and co-authored the Comprehensive Mental Health Support Act, securing state senate co-sponsorship

New Jersey Privacy Movement | *Founder*

Jan 2024 – Mar 2024

- Authored a public petition for user data rights and collaborated with NJ State Senator Turner to draft a formal privacy bill

FBLA | *7th Place Regional Champion*

Sep 2022 – June 2026

- Ranked 7th in Computer Problem Solving for Central NJ; raised \$150+ and collected 100+ shoes for charity drives